

Tests and Trials Purpose and Scope

Tests and Trials are important components of an incremental approach to building the new Environmental Land Management scheme.

We began the Tests and Trials work in 2018 as a vehicle to engage with a wide range of farmers, land managers and stakeholders to harness their ideas to conduct research to contribute to the design of the new Environmental Land Management scheme. Farmers and land managers taking part in the Tests and Trials co-design elements or building blocks of the new scheme in their real-life environment. We use this evidence and learning they generate to understand farmer and market behaviour and to help shape and inform scheme design.

Tests and Trials are a safe space to fail, learn and improve. As well as Testing the key building blocks of the new scheme, we are also Trialling new ideas and innovative approaches to understand if and how they could be built into the new scheme and help to shape the national pilot.

Tests and Trials and the National Pilot are important components of an incremental approach to building the new scheme. The learning they generate will help us build a tried and Tested framework for the wider roll out of the new scheme. This learning and improving won't stop there as we will continue testing until 2028, with the overall aim that in 10 years' time, we will have a tried and Tested 'co-designed' system that is world-class.

The scope of Tests and Trials is outlined below:

- Co-design and Test how potential **elements of the new scheme work in a real-life environment, with different user groups, across different geographies**
- **Trial new or innovative approaches** that we haven't previously utilised to understand if and how they could be used within a new scheme
- Provide **real time evidence** to help shape policy and delivery
- A means to **engage with a wide range of farmers and land managers** to help **prepare the sector for change** and **harness ideas and delivery solutions**
- It **will not assess** if specific interventions or delivery practices achieve **environmental outcomes**

We are using a **phased approach** to deliver Tests and Trials inviting proposals from stakeholders, this call represents our third phase with Tests and Trials due to continue until 2028, supporting the design and implementation of the Environmental Land Management scheme. This approach has enabled us to prioritise Testing the building blocks as we analyse and evaluate the results from Tests and Trials to feed into our policy development process.

Call Purpose and Scope

To help the development of the new Environmental Land Management Scheme(s) we are conducting a variety of Tests and Trials, aimed at conducting research to contribute to our understanding of how parts of the new scheme work in a real-life environment.

We are working with a range of different stakeholders, across different geographies to undertake the Tests and Trials, including farmer groups, representative bodies and non-governmental organisations.

The most ambitious component within the Environmental Land Management scheme(s) is Landscape Recovery. Landscape Recovery will support the restoration and enhancement of natural ecosystem function at scale through transformative land use change projects delivered over the long term. Such projects will restore wilder landscapes in places where this is appropriate, through the creation of woodland, restoring wetlands and peatlands, enhancing bogs, fens or saltmarshes or a mosaic of different habitats. These projects will **contribute towards Environmental Land Management environmental objectives** and support local environmental priorities while making an important contribution to national targets. We expect that this component will be competitive.

Tests and Trials are inviting proposals from stakeholders to Test and Trial the mechanisms required to deliver long-term land use changes with farmers and land managers, for example Testing the practical elements of long-term commitments (30+ years), developing models that can be rolled out more widely and Trialling innovative mechanisms such as conservation covenants. Tests and Trials should be designed to be able to make policy and delivery recommendations that are suitable for roll out within landscape recovery throughout England, rather than recommending approaches that are only applicable in specific circumstances or locations. The specific areas of interest identified for this call are outlined below:

- a. How to balance the delivery of national and local environmental priorities in selection and design of projects (e.g. develop and Test models that identifies national opportunities aligned with local priorities and the role of individuals, groups and organisations in balancing local delivery with national priorities);
- b. How to construct agreements for different land ownership structures e.g. individual and group agreements, tenants, commons
- c. How to blend public and private finance in funding projects (e.g. identifying and Testing innovative mechanisms to use public investments to leverage private sector funding);
- d. How to align Landscape Recovery projects (as distinct from Local Nature Recovery projects) to wider Defra initiatives such as Biodiversity Net Gain and Nature for Climate projects which may also deliver land use change (e.g. Testing the practicalities of differentiating between work delivered under wider Defra initiatives and actions and outcomes which might be delivered under an Environmental Land Management scheme agreement).
- e. How to incentivise land manager participation and collaboration in landscape recovery projects and determine appropriate payment mechanisms (e.g. identify and Test the measures which would encourage participation, the level of incentive required and the mechanisms for facilitating incentivisation);
- f. How to construct long-term agreements (30+ years), potentially incorporating conservation covenants, to safeguard investments in land use change and associated environmental outcomes (e.g. Test what information might be required to deliver and safeguard long term commitments to land use change);

Through Tests and Trials, we will test and trial the mechanics of how Landscape Recovery could work and make policy and delivery recommendations that can be built into the National Pilot. Over the next four years through the National Pilot we will establish 10 Landscape Recovery projects to help restore wilder landscapes, with a focus on large-scale sites. Later this year we will publish more details on how we will choose early pilot projects and invite nominations of potential sites. Together, the National Pilot and Tests and Trials will provide us with a solid basis for testing the delivery capability and implementation of Landscape Recovery, which in turn will help inform the full roll out of Environmental Land Management in 2024.

Tests and Trials are a safe space to fail, learn and improve, they allow us to look at new and innovative approaches that we haven't tried previously to help us understand how they could be used within the new scheme. For example, we are Testing the Payments by Results approach with farmers in Yorkshire and East Anglia. The Trial will examine how the behaviour of farmers changes as they take greater responsibility for delivery of the outcomes, Testing the accuracy of self-assessment and the potential role of remote sensing in validating results.

By Testing and Trialling elements of the new scheme, Defra is looking to work together with farmers and land managers and other stakeholders to harness their ideas, gain their feedback and build something that works for the diverse needs of the agriculture sector, whilst also improving our environment.

We have used a **phased approach** to deliver Tests and Trials inviting proposals from stakeholders. This has enabled us to prioritise Testing that will help prepare for the National Pilot and 2024 scheme design. We have Tests and Trials that cover 7 areas or themes:

- **Land Management Plan** – what would be included in a plan, how long it should be and what information is needed to support the land manager or farmer
- **Role of advice and guidance** – the level and role of advice and guidance land managers and farmers would need to put together a plan and implement it effectively
- **Payments** – to Test different approaches to and how these might work in practice
- **Spatial prioritisation** – to Test how different spatial mechanisms identify and agree local priorities
- **Collaboration** – to Test how different methods of collaboration work to deliver environmental outcomes on a greater scale
- **Innovative delivery mechanisms** – to Test how innovative solutions work on the ground and in what circumstances. Examples include reverse auctions and payment by results

Strategy – Testing the different models, mechanisms and approaches to achieve ambitious land use change

Help with completing a taxonomic proposal to ICTV

ICTV welcomes taxonomic proposals from any interested individual. Please complete the relevant template modules (see Instructions, below) and e-mail them to the appropriate Subcommittee Chair. Information on the seven virus Subcommittees and their Chairs can be found at <https://ictv.global/sc>.

Proposals will be forwarded to all relevant Study Groups and comments relayed to the authors. The proposals are also available for public comment on the ICTV web site until the next annual meeting of the Executive Committee (EC), when Subcommittee Chairs (or their deputies) present the taxonomic proposals for discussion. Straightforward proposals to create new species in an existing genus or to create new genera in an existing family may be approved at the next EC meeting. Some proposals, especially more complex proposals, may be approved conditionally and will continue to be available for public comment before a final decision by the EC is reached. Importantly, proposals do not become accepted taxonomy until a final ratification vote by the ICTV membership is conducted. The ratification vote usually occurs only once annually in February.

1. The information needed by ICTV

Carefully completed proposals are not only easier for your colleagues to assess but are also more likely to proceed smoothly.

- Please provide enough information with any proposal for the creation of a new taxon (species, genus, etc.) to show that it has the **required properties** to be placed in any **more highly ranked taxon** (genus, family, order, etc.) to which it is being assigned and that it is sufficiently **distinct from its most closely related taxon**.
- Proposals for new species will need to state the **species demarcation criteria** established for the genus and **show that the candidate species meets these requirements**. Each new species must be assigned to a genus and a family, whether existing or novel.

- Proposals for the creation of a new taxon of higher order require clear justification with reference to all related existing taxa. The ICTV provides for 15 levels of taxonomic classification (from realm to species) but, when proposing a new taxon, you need only to refer to its placement within the next highest existing taxonomic level.
- Most proposals benefit from a **phylogenetic tree**: please provide a tree where **branch length is proportional to genetic distance**, generated using an appropriate algorithm (e.g., Maximum Likelihood or Maximum Clade Credibility) and provide evidence of the reliability of the branching (e.g., by bootstrapping, posterior probability, aLRT). Use colour differences, symbols etc., to draw attention to the position(s) of the proposed new taxa and provide a legend or key so that existing taxa can easily be recognised. Tip labels should be annotated with INSDG nucleotide accession numbers but individual sequences or clades should be also annotated with virus or taxon names. Trees can often be converted easily to something much more informative (see <http://veb.tumc.nl/SNAD>).
- Proposals are posted on the ICTV public web site, so please be aware that copying content from publications requires an appropriate **acknowledgement** and may require prior permission from the **copyright holder**.
- If a proposal involves a **new taxon name that has been derived from the name of a living person**, you are required to provide evidence that the person has authorised use of their name, and that they agree to the form in which **their name or an element of it** is to be used. Evidence of authorisation may be in the form of a signed letter or email which must be included as an attachment to the proposal. These attachments will not be posted online but will be retained by the ICTV. It is ICTV practice not to name the realms after persons, living or deceased.

2.—Taxonomic proposal templates

Proposals must be prepared using the ICTV templates. As these templates may change periodically, you must use the most recent versions which are available at <https://ictv.global/taxonomy/templates>. There are two templates:

- A Word-formatted module (**TP_Template_Word_module_2024**) for title, authors, approvals and supporting evidence. It also contains a section for non-standard proposals (see §3, below).
- An Excel module (**TP_Template_Excel_module_2024**). This is used to present the proposed taxonomic change(s) as a comparison of old and new taxonomic structures. This is described in more detail in §5 below.

3.—Types of proposal

Taxonomic proposals: The great majority of proposals will be to create new taxa and/or change existing taxa. For all such proposals you will need to complete **BOTH** Parts 1 and 3 of the Word module (**TP_Template_Word_module_2024**) **AND** the Excel module, (**TP_Template_Excel_module_2024**). Please follow the instructions shown in **red font**.

NOTE: The **Excel module is the critical document** that will be used to implement the proposed taxonomic changes once they are approved and ratified. If the information required in the Excel module is presented only in the Word module, **the taxonomic changes cannot proceed**.

Non-standard proposals: For any proposal regarding ICTV procedures, rules or policy, not involving the creation of new taxonomy, it is usually necessary to complete only Parts 1 and 2 of **TP_Template_Word_module_2024.docx**. The Excel module is not normally needed.

4.—Authorship of proposals

ICTV will publish lists of proposals when they are ratified and will cite the proposal in standard *Archives of Virology* format with a list of authors, a short title incorporating the Code assigned for tracking purposes, and the web link to the ICTV site. Please ensure that all authors are listed in the desired order with a 4-part affiliation (department, institution, city and country) and email address for **each author** in the Table provided. Add additional rows as required. Please also indicate **a single corresponding author**. Complete and correctly formatted author information is essential for future processing. Incorrectly filled forms will be returned to the submitter.

5.—Steps in approval of taxonomy proposals by the ICTV Executive committee (EC)

ICTV will publish lists of proposals when they are ratified and will cite the proposal in standard *Archives of Virology*

6.—How to present taxonomic changes in the spreadsheet

Guidelines for completing the Excel module (TP_Template_Excel_module_2024) are provided below. A number of simple and more complex worked examples are provided in the document **Taxonomic_Proposals_Examples_2024** which is also available at <https://ictv.global/taxonomy/templates>.

Proposed taxonomic changes are to be presented by reference to old and new taxonomic structures—old on the left, new on the right—each set out hierarchically as in the Master Species List (<https://ictv.global/msl>). Please follow the instructions in blue at the top of the spreadsheet.

General rules

- Existing taxonomy hierarchy (columns A–O) is compared with that proposed (Columns P–AK). Each row of the spreadsheet conveys information about only one taxon; i.e., every taxon that is subject to change has its own, dedicated row in the spreadsheet.
- The subject of each row is the lowest ranked entry on that row (i.e., the entry furthest to the right). The other entries on that row provide context, specifying the old and new taxonomic positions.
- When a new taxon is created, the left-hand side of the row (columns A–O) is left blank. Conversely, when a taxon is abolished, the right-hand side (columns P–AK) is blank. Columns AL and AM, which specify the taxonomic change, must be filled.
- Any change to a higher taxon (assign, move, abolish, etc.) is deemed to apply also to its constituent taxa, unless action is taken to move them elsewhere.

For each new species, you should provide:

- Column AE, **Exemplar**: the Genbank Accession Number for the exemplar virus referred to in column AG. This entry is **mandatory**.
- Column AF, **Exemplar**: the name of the exemplar virus referred to in column AF. Please use the correct orthography (see below).
- Column AG, **Abbreviation**: If you wish to define an abbreviation for the virus name, enter it here.
- Column AH, **Isolate designation**: A short code (usually obtained from the GenBank record) used to identify the specific isolate of the exemplar virus (enter "nk" if not known).
- Column AI, **Genome coverage**: Please indicate the degree of completeness of the sequence (CG—complete genome; CGG—complete coding genome; PG—partial genome) using the drop-down menu. The exemplar sequence will normally be expected to be a complete genome and must be publicly available.
- Column AJ, **Genome composition**: Please indicate the genome composition of the viruses that are being proposed in the new taxon using the drop-down menu.
- Column AK, **Host/Source**: Please indicate the host of the virus or source of the genomic sequence from environmental samples using the drop-down menu.

- The 15 taxonomic ranks listed in TP_Template_Excel_module_2024.xlsx are the ones currently used in ICTV taxonomy. It is not possible to make a taxonomic proposal dealing with "tentative species" or any category below the rank of species.

- Guidelines for naming all taxonomic ranks (i.e., from realm to species) are set out in the ICTV Code of Virus Classification. Rules on orthography of virus species, other taxon names and virus names can be found in the document named [How_to_write_virus_species_and_other_taxa_names.pdf](https://ictv.global/files/info), available at <https://ictv.global/files/info>.

Notes:

1. Virus names, abbreviations, isolate designations and the virus host/source (columns AF, AG, AH and AK) have no official taxonomic status and are not a formal part of the proposal. However, they will be listed in the ICTV Report and in summaries of taxon information such as the Virus Metadata Resource (<https://ictv.global/vmr>). Comments in column AN serve as an aid to readers and are not a formal part of the proposal.
2. Names of **new taxa** (including renamed taxa) should be written in **red font** for clarity. This should **not** be done for taxa that already exist or are only moved.

~~4. Multiple entries in column AE (Exemplar GenBank Accession Number) should be separated by a semicolon delimiter. This applies, for example, to virus genomes that are segmented.~~

- ~~• If you have any further questions about the forms or the procedures, please contact one of the Secretaries or Subcommittee Chairs. Names and e-mail addresses can be found at <https://ictv.global/members/ec-members>.~~